

Gabriele La Malfa

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While studying for my master's degree in accounting, I got hugely passionate about game theory. As I continued my business studies, I focused on moving toward science. I learned programming, mathematics, and statistics on my own, which helped me gain admission to the PhD program in computer science at King's College London. There, I was able to fully pursue my passion for science. Now, in my final year of doctoral studies, I am researching multi-agent reinforcement learning, as well as the safety and fairness of AI. Lately, my work has focused on self-play, large language models, the Theory of Space, and vision-language models.

I am a human-oriented person. Projects are exciting, but people around me and collaborators always come first.

ACADEMIC POSITIONS

PhD candidate in [Safe & Trusted AI](#) at King's College London & Imperial College London Oct. 2022 – Current
Safety and Fairness in Multi-Agent Systems.

– Supervisors: Prof. Elizabeth Black, Dr. Jie M. Zhang and Prof. Michael Luck.

Visiting Researcher at [Nokia Bell Labs](#) Jul. 2025 – Dec 2025
Classification of AI agent risks in the workplace.

– Researching AI impacts on society under the supervision of Prof. Daniele Quercia.

Research Assistant at the University of Cambridge, [Centre for Risk Studies](#) Oct. 2020 – Aug. 2022
Research and modelling business, financial and cyber risks.

– Co-published: '[Mitigating ransomware risk: determining optimal strategies for businesses.](#)'

WORK EXPERIENCE

Quantitative Risk Officer Jun. 2020 – Aug. 2020
Deutsche Börse AG

EDUCATION

EMLyon Business School 2020
MSc Quantitative Finance

University of Eastern Piedmont 2017
Master's Degree in Accounting and Business Administration

PUBLICATIONS

[The Attacker in the Mirror: Breaking Self-Consistency in Safety via Anchored Bipolicy Self-Play.](#)
arXiv. **Gabriele La Malfa**, Emanuele La Malfa, Saar Cohen, Jie M. Zhang, Michael Luck, Michael Wooldridge, Elizabeth Black.

[Fairness Aware Reinforcement Learning via Proximal Policy Optimization.](#)
AAAI Conference on Artificial Intelligence (AAAI-26). **Gabriele La Malfa**, Jie M. Zhang, Michael Luck, Elizabeth Black.

[Large Language Models Miss the Multi-Agent Mark.](#)
NeurIPS 2025, Position Paper Track. **Gabriele La Malfa**, Emanuele La Malfa, Samuele Marro, Jie M. Zhang, Elizabeth Black, Michael Luck, Philip Torr, Michael Wooldridge

[Using Protected Attributes to Consider Fairness in Multi-Agent Systems.](#)
ECAI 2024, AEQUITAS Workshop. **Gabriele La Malfa**, Jie M. Zhang, Michael Luck, Elizabeth Black.

[Deep Neural Networks via Complex Network Theory: a Perspective.](#)
IJCAI 2024, Main Track. **Gabriele La Malfa**, Emanuele La Malfa, Giuseppe Nicosia, Vito Latora.

CURRENT PROJECTS

Self-play red teaming avoiding telepathy. I am currently developing new approaches for self-play red teaming to enhance the safety of LLMs in a computationally efficient way. My research involves analysing the Nash equilibria of the game and proposing a method that prevents telepathy between the attacker and defender.

Multi-agent Theory of Space with deception of focal points. I am exploring the Theory of Space, i.e., how embedded VLMs build through active exploration an internal map of the surrounding environment in the presence of deception of focal points and coordination.

TEACHING

Artificial Intelligence <i>Practical</i>	2023–2025
Mathematics and Statistics for AI <i>Group Tutorial Lead</i>	2024–2025
Machine Learning <i>Group Tutorial Lead</i>	2023–2024
Logic and Knowledge Representation <i>Group Tutorial Lead</i>	2023–2024
Elementary Logic and Applications <i>Group Tutorial Lead</i>	2023–2024
Big Data Technology <i>Group Tutorial Lead</i>	2023–2024

SERVICE & ORGANISATIONS

LOD Conference Series (2020–2025) <i>Organizing Committee</i>
AAAI Conference <i>Reviewer</i>

VOLUNTEERING

I am committed to social justice, ranging from dialogue leadership at KCL to humanitarian support for students and people who are not as lucky as I am.

Lunchtime Dialogues: Identity & Social Justice <i>Lead Facilitator: Organised a dialogue series across all King's College London faculties to raise awareness on social justice and human rights issues.</i>	2023 – 2025
Ukrainian Global University <i>Interviewer: Supported the international placement program by conducting admissions interviews for Ukrainian students seeking to study at partner universities abroad.</i>	2022 – 2023
The Kharkiv and Przemyśl Project <i>Volunteer: Provided on-the-ground support in Przemyśl, Poland, assisting refugees arriving from and returning to Ukraine.</i>	August 2022
Cambridge University Amnesty International (CUAI) <i>Executive Committee Member: Led local campaigns including Write for Rights (W4R) and various global human rights initiatives.</i>	2021
Eutanasia Legale <i>Volunteer: Coordinated signature collection for a national referendum to decriminalise euthanasia in Italy, contributing to a total of 1.2 million signatures.</i>	July 2021
Centro Gazza Ladra <i>Project Organiser: Managed crowdfunding and event logistics for an association dedicated to supporting children with special needs.</i>	2018